



Landscape fragmentation and population distribution in refugee camps: evidence from the Rohingya refugee influx in Bangladesh

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Abstract

This study investigates landscape changes after the 2017 Rohingya migration from Myanmar to Bangladesh. Land-use and land-cover (LULC) classification techniques and fragmentation analysis were used to assess the changes in the spatial structure of the landscape between 2016 and 2024. Additionally, analysis of changes in the Normalized Difference Vegetation Index (NDVI) was employed to identify forest degradation. Camp-specific changes in forest and settlement areas were also observed. Sentinel-2 images were used to classify the camp areas into four classes: forest, water, agriculture/open field, and settlement. To quantify landscape fragmentation, seven class-level and two landscape-level metrics were utilized. The results indicated a 54.6% decline in total forest area from 1688.8 to 767.3 hectares, while settlement areas expanded from 144.8 to 1011.5 hectares (a 598.5% increase). Both class-level and landscape-level metrics revealed increased landscape fragmentation in 2024 and the analysis of NDVI identified the remaining patches of healthy forest inside the study area. Correlation analysis between forest and settlement change showed a strong negative correlation, demonstrating the connection between rapid migration and forest loss. The most populated Camp (15), with approximately 57,922 refugees, had the greatest increase of 67.1 hectares in settlements and decrease of 62.8 hectares in forest. Forest area increased in the Kutupalong and Nayapara camps and Camps 23, 24, and 25. Findings from this study can help optimize relocation plans, strengthen forest protection, and support camp restoration efforts.

Keywords Refugee · Migration · Forest · Fragmentation · LULC · Rohingya · Bangladesh

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Introduction

The Rohingya humanitarian crisis is a long-standing problem that has evolved through various phases over decades, resulting in the forced migration of Rohingya people to neighboring countries including Bangladesh. This persistent crisis is rooted in the systematic discrimination and violence perpetrated by the Government of Myanmar against the Rohingya population (Ragland, 1994; US Department of State, 2022). The crisis peaked around 2017 when over 600,000 Rohingyas crossed the Bangladesh-Myanmar border to escape repression and seek refuge in Bangladesh (Assessment Capacities Project (ACAPS), 2017; Kader & Choudhury, 2019; United Nations High Commissioner for Refugees (UNHCR), 2018). This marked the largest influx of Rohingya refugees into Bangladesh (ACAPS, 2017; Kader & Choudhury, 2019; UNHCR, 2018). As of April 30, 2024, approximately 979,306 displaced Rohingya individuals are residing in the refugee camps in Bangladesh, a figure that excludes the undocumented Rohingya settlers who have assimilated with local communities outside the camps (Human Rights Watch (HRW), 2000; Idrish & Khatun, 2018; UNHCR & Government of Bangladesh (GoB), 2024a). As a result, Bangladesh now hosts the largest stateless population globally (World Population Review, 2024).

The sudden influx of Rohingya refugees in 2017 presented significant challenges for Bangladesh. In 2016, the country, with an area of 147,570 km², had a population of approximately 160 million. A full 1.5 million residents live below the international poverty line (World Bank, 2023, 2024a). Being a lower middle-income country with socio-economic and climate vulnerabilities, Bangladesh did not have enough resources to manage the substantial number of refugees (Shakhawat et al., 2020; UNHCR, 2018; World Bank, 2024b). Therefore, the country sought emergency resources and assistance from international humanitarian organizations and allocated its forest reserves to establish refugee camps (Paul, 2017; UNHCR, 2018). Refugees arrived before any infrastructure was built at allocated campsites, leading to the spontaneous growth of settlements in pre-existing camps and the informal development of new extensions (UNHCR, 2018). Therefore, the rapid construction of camp infrastructure to provide essential services (e.g., shelter, access to food and water, sanitation) was executed without adequate planning (UNHCR, 2018), and the local environment, with abundant forest, was quickly transformed (UNHCR, 2018). Significant deforestation and degradation of forest land associated with establishing campsites, as well as the use of forest resources by refugees for camp activities, have been confirmed by several studies (Hasan et al., 2021; United Nations Development Programme (UNDP) Bangladesh and United Nations (UN) Women Bangladesh, 2018). This rapid transformation has had detrimental effects on the local environment, including increasing surface temperature, soil erosion, and pollution, leading to overall ecosystem degradation (Bappa et al., 2022; Kamal, 2022; Rashid et al., 2021; UNDP Bangladesh and UN Women Bangladesh, 2018).

Globally, the impact of refugee influx is often assessed by land-use and land-cover (LULC) classification (Al Shogoor et al., 2022; Gromny et al., 2024).

Gromny et al. (2024) studied the land-cover changes in the Mtendeli refugee camp of Tanzania between 2016 and 2022 and found decrease in vegetation and increase in artificial surfaces after the camps were established. Al Shogoor et al. (2022) observed expansion of urban and agricultural land and decline of rangelands and bare lands in Northwestern Jordan following the Syrian refugee migration in 2011. In Bangladesh, several studies have employed LULC classification techniques to evaluate the impact of the Rohingya influx (Bappa et al., 2022; Hassan et al., 2023; Hossain & Moniruzzaman, 2021; Sakamoto et al., 2021; Sarkar et al., 2022). Hassan et al. (2023) investigated long-term regional forest cover changes (1989–2015) in the Teknaf peninsula (Ukhia and Teknaf sub-districts) of Bangladesh using a cart regression model for LULC classification revealing periodic gains and losses of forest over the years. They found significant loss of both protected and non-protected forest after the 2017 refugee influx followed by an increase in both categories in 2019 and 2021 due to restoration efforts. In addition, Hassan et al. (2023) conducted a fragmentation analysis to observe the spatial arrangement of forest cover; however, detailed fragmentation within camp areas for all the LULC classes was not explored in that study. Hossain and Moniruzzaman (2021) also investigated forest degradation in this area using both LULC classification and Normalized Difference Vegetation Index (NDVI). Unlike these regional analyses, Bappa et al. (2022) focused entirely on LULC changes in the Kutupalong-Balukhali site located in Ukhia sub-district of Cox's Bazar district, where the majority of the Rohingya population resides. They used the maximum likelihood classification technique to classify the site into four LULC classes: forest, settlement, bare land, and wetland. Their study showed an approximately 74% decrease in forest areas between 2015 and 2021, with forest areas shrinking from 1256.55 hectares in 2015 to 432.61 hectares by 2018 and further declining to 321.32 hectares by 2021.

Several research gaps have emerged in the existing studies on the Rohingya influx. While many have analyzed LULC changes, most have focused on the broader region (i.e., the Teknaf peninsula), with limited attention to the specific changes within refugee camps and their current development. Similarly, Hassan et al. (2018) and Hossain and Moniruzzaman (2021) analyzed changes in NDVI to observe forest degradation in a regional context and did not elaborate on the camp areas. Hassan et al. (2023) conducted fragmentation analysis primarily on forest cover, but not other land-use classes. The relationship between population size in individual camps and its impact on LULC changes and forest loss also remains unexplored. Lastly, with the ongoing relocation of Rohingya populations from the original camps to Bhasan Char, the government-allocated island in Noakhali district, a current assessment of LULC and population dynamics in the camps is needed to inform future restoration of these areas.

The goal of this study is to provide a comprehensive understanding of Rohingya refugee migration and land transformation in Bangladesh during the large-scale influx of this population in 2017. The specific objectives are to determine the recent, post-influx land uses and LULC changes following the 2017 migration, analyze patterns of landscape fragmentation with an emphasis on forest fragmentation, investigate forest degradation, and observe the relationship between population influx and

forest loss within camps. By combining LULC classification, NDVI, fragmentation, and population distribution analyses, this research offers a more detailed understanding of the land-use changes within the refugee camps and their implications for the environment. However, examining the impact of Rohingya migration without an adequate understanding of the ongoing humanitarian crisis poses a significant constraint on achieving a comprehensive understanding of this complex situation and the long-lasting effects triggered by the 2017 Rohingya influx. Therefore, a historical overview of the Rohingya refugee crisis is provided prior to the analysis.

Historical background on the Rohingya refugee crisis in Bangladesh

The Rohingyas belong to a distinct ethnic group from the Rakhine state of Myanmar. They are minority among the predominantly Buddhist and other Rakhine populations of Myanmar and distinguished by their Muslim faith and Bengali (rather than Burmese or Rakhine) linguistic dialect (Ragland, 1994). The Rakhine state, formerly known as Arakan state, is located along the western coast of the Bay of Bengal and shares a 180 km border with the Cox's Bazar district of Bangladesh. Sources indicate multiple instances of Rohingya displacement across the Bangladesh-Myanmar border as a result of systematic oppression, ethnic discrimination, violence, and their eventual exclusion from Myanmar citizenship (Kader & Choudhury, 2019).

Although the Rohingya's presence in Myanmar goes back to the twelfth century, the Government of Myanmar refused the Rohingya's right to citizenship and classified them as foreigners during the 1977 Naga Min census (ACAPS, 2017; HRW, 2000; Kamal, 2022; Ragland, 1994). Consequently over 200,000 Rohingya fled to Bangladesh in 1978 to escape from forcible evictions through military violence and brutality (HRW, 2000). The UN assisted the overwhelmed Government of Bangladesh in establishing fourteen temporary camps along the border (ACAPS, 2017; HRW, 2000). Thirteen camps were located in Cox's Bazar, while one camp was located in Bandarban (ACAPS, 2017; HRW, 2000). Negotiations between the two governments facilitated the repatriation of about 180,000 Rohingya to Myanmar during 1978–1979 (ACAPS, 2017).

The Emergency Emigration Act of 1974, followed by the Citizen Act of 1982, and lastly the exclusion of the Rohingya population under the Citizenship Act of 1989 finalized the “stateless” status of the Rohingya population in Myanmar (Kader & Choudhury, 2019; Kamal, 2022). In 1991–1992, increased military repression, forced labor, torture, rape, killings, and other violence prompted another influx of approximately 250,000 Rohingyas into Bangladesh, who took shelter in nineteen camps near Cox's Bazar with support from UNHCR and NGOs (ACAPS, 2017; HRW, 2000; Kader & Choudhury, 2019; Kamal, 2022). The Kutupalong and Naya-para camps that exist at present were established as the official registered camps at that time, specifically for registered refugees. The rest of the camps were considered as temporary establishments near the registered camps (ACAPS, 2017). During this period, the Government of Bangladesh emphasized repatriation, rejecting any possibility for local integration (ACAPS, 2017). Consequently, new arrivals after the initiation of the repatriation process were denied access to registered camps and left

in a state of undocumented status. This situation complicated the documentation of the subsequent influx of 200,000–250,000 Rohingyas in 1997, who fled Myanmar to escape economic hardship and forced labor. Sources indicate that the undocumented Rohingyas either settled in makeshift camps or local Bangladeshi villages (ACAPS, 2017; Kader & Choudhury, 2019). Idrish and Khatun (2018) found that similarities in language and religion facilitated the assimilation of Rohingyas into the local host communities. Between 1993 and 1997, approximately 230,000 Rohingyas were repatriated to Myanmar under a memorandum of understanding between Bangladesh and the UNHCR (ACAPS, 2017).

The last and most substantial migration occurred in 2016–2017, with over 87,000 fleeing in 2016 due to military repression, followed by approximately 691,768 additional individuals in August 2017 (UNHCR, 2018). While the military claimed their violent actions were in response to border attacks from the insurgence group known as the Arakan Salvation Army (ARSA), reports suggested these measures were excessively brutal, coordinated, and intended to terrorize and displace the Rohingya population (Kader & Choudhury, 2019; US Department of State, 2022). The mistrust in the Government of Myanmar and the fear of reprisals have continued to distress the refugees, one of several factors impeding repatriation efforts (Rahman, 2023; UNHCR, 2023).

Research objective

The research objectives of this study are (1) to determine the LULC changes and land fragmentation in refugee camps between 2016 and 2024, (2) to observe forest degradation following 2017 influx, and (3) to assess the relationship between population influx and forest loss within camps.

Methodology

Study area

Bangladesh is located in a low-elevation deltaic plain north of the Bay of Bengal in South Asia, sharing a border with India to the west, north, and north-east and with Myanmar to south-east. This study is concentrated on the Rohingya refugee camps comprising an area of approximately 2500 hectares (Fig. 1) in Ukhia and Teknaf sub-district of Cox's Bazar district of Bangladesh. These two sub-districts are part of the Teknaf peninsula, located in the southeastern region of Bangladesh, where they share an international border with Myanmar. This area has a subtropical monsoon climate with an annual average temperature of 26.1 °C and annual average rainfall of 4000 mm, 80% of which occurs in the monsoon season from June to August (Bappa et al., 2022; Hassan et al., 2023). It comprises a diverse landscape including hills, tidal floodplains, piedmont plains, and beaches (Moslehuddin et al., 2018). The maximum elevation of the piedmont plains is 10 m (Hassan et al., 2018). Current land uses in this area include agriculture, aquaculture, salt farming, fishing, human

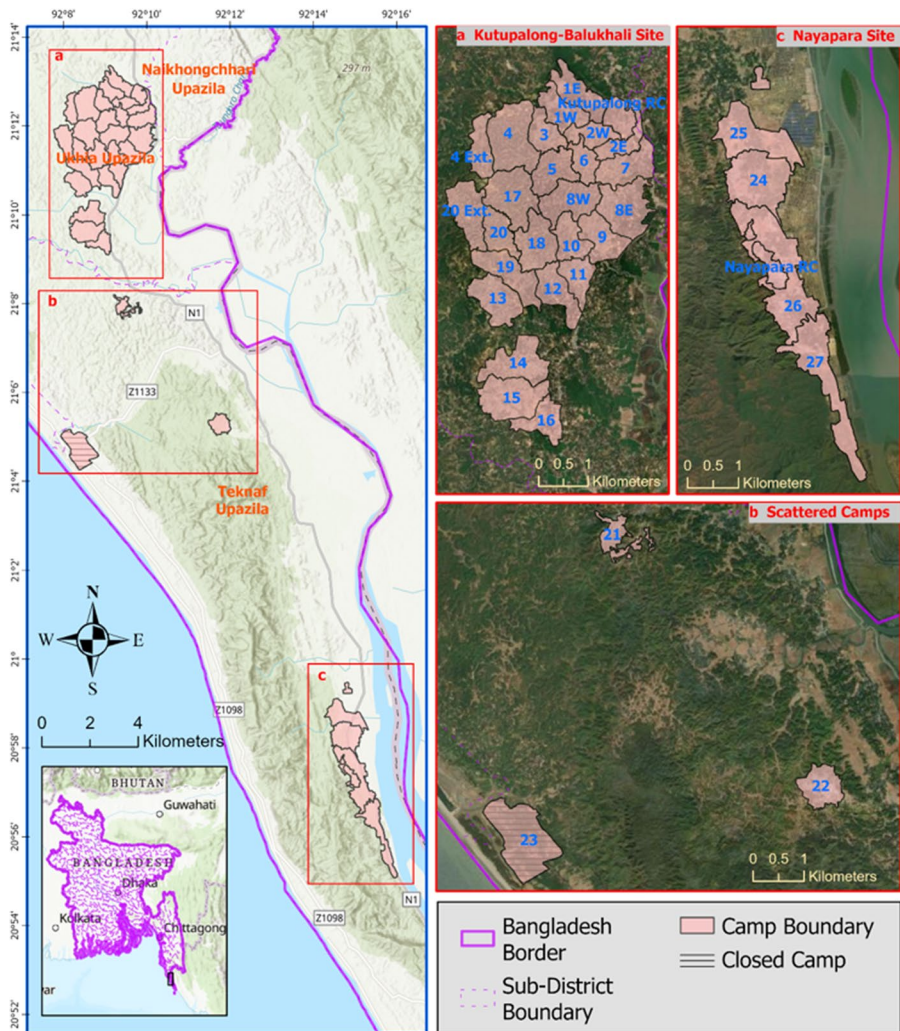


Fig. 1 Map of Rohingya refugee camps in Cox's Bazar district: **a)** Kutupalong-Balukhali site **b)** Scattered camps, and **c)** Nayapara site

settlement, and tourism facilities (UNDP Bangladesh and UN Women, 2018). The Kutupalong-Balukhali site and Nayapara expansion site are situated at the eastern edge of Inani national park and Teknaf wildlife sanctuary, respectively, both protected forest areas recognized for their rich biodiversity and declared critical habitats for wildlife. UNDP Bangladesh and UN Women (2018) reported significant degradation and habitat loss in the protected forest areas due to the Rohingya 2017 influx.

There are 35 Rohingya refugee camps in Bangladesh, among which 34 are in Cox's Bazar district (the other is in Bhasan Char of Noakhali district). However, Camp 23 of Cox's Bazar was closed in 2021 (Haque, 2021). These 34 camps are

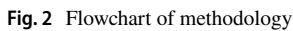
situated between 20° 55' N to 21° 14' N latitude and 92° 7' E to 92° 17' E longitude. Most refugee camps (26 camps) are clustered in the Kutupalong-Balukhali site located in the Ukhia sub-district with additional camps dispersed across nearby Hakimpura (Camp 14), Jamtoli (Camp 15), and Bagghona/Potibonia (Camp 16). Some additional, scattered camps are located south Kutupalong-Balukhali site at Chakmakul (Camp 21), Shamlapur (Camp 23), and Unchiprang (Camp 22). The second largest camp cluster (5 camps) is near Nayapara registered camp located in the Teknaf sub-district, and this includes camps from Alikhali, Leda, and Jadimura (UNHCR, 2018). Some camps were established around existing local households, especially in the Nayapara site, where refugees and locals live in close proximity. As of April 30, 2024, there are 961,729 registered Rohingya refugees in Bangladesh, with 944,154 of them residing in Cox's Bazar and the remainder (35,152 registered individuals) living in the Government of Bangladesh allocated island, Bhasan Char of the Noakhali district (UNHCR & GoB, 2024b). Relocation to this island started in 2020 (Beech, 2020). Only the 34 refugee camps located inside the district of Cox's Bazar, excluding Bhasan Char, are considered as the study area (Fig. 1).

Data collection and processing

This study utilized Sentinel-2A products from 30 November 2016 and 03 January 2024. These products are radiometrically and geometrically corrected, include cloud masks, have 13 bands with 10, 20, and 60 m spectral resolution, and have been freely available since June 2015 (European Space Agency, 2015). These two images were chosen to capture scenarios before (2016) and after (2024) the migration. Data from the winter were selected to ensure good quality images with low cloud cover. Following the methodology of Hassan et al. (2018), we avoided using images from the rainy season to minimize inaccuracies in NDVI analysis caused by vegetation's sensitivity to rainfall. Bands 2 (490 nm), 3 (560 nm), 4 (665 nm), and 8 (842 nm) of the Sentinel-2 satellite capture the blue, green, red, and near-infrared wavelengths of the electromagnetic spectrum (European Space Agency, 2015). These bands offer the highest resolution (10 m) and, therefore, were used for generating false-color composite images to highlight different features of LULC classes, which facilitated the selection of the training samples. A shapefile of the study area was collected from the Humanitarian Response (2023) website and projected with the WGS 1984 UTM Zone 46N projection system. The classification analysis was performed in Google Earth Engine (GEE).

LULC classification procedures and accuracy assessment methods

The LULC classification was performed on the Sentinel-2 images as outlined in Fig. 2 following the approach by Sarkar et al. (2022). Images were classified into four classes: forest, water, agriculture/open field, and settlement. The forest class comprises natural and homestead forests, such as hill forests, orchards, rubber plantations, bamboo forests, and mangrove plantations. The settlement class includes all



Following the studies of Nasiri et al. (2022), Talukdar et al. (2020), and Waśniewski et al. (2020), this study utilized the Random Forest (RF) machine learning technique for LULC classification. In this study, 2000 sampling points (including training and validation samples) were obtained in GEE using manual interpretation and high-resolution GEE images. The samples were randomly divided into two sets: approximately 80% of these samples were allocated for model development, and the remainder were used for model validation (Table 1). A study by Nasiri et al. (2022) applied a similar method to classify and validate Landsat-8 and Sentinel-2

Year	LULC class	Training	Validation
2016	Water	396	104
	Forest	405	95
	Agriculture/open field	378	122
	Settlement	396	104
2024	Water	404	96
	Forest	404	96
	Agriculture/open field	399	101
	Settlement	400	100

images in GEE, where they divided their samples into training (60%) and validation datasets (40%).

The non-parametric RF model is the combination of random decision trees developed by Breiman (2001). In recent times, this method has been widely used for both classification and regression. Some of the important parameters for developing the RF model include the number of decision trees (ntree), the number of features per split (mtry), randomization seeds, the minimum number of leaf populations, and the maximum number of leaf nodes in each tree. As the number of trees increases, the overall accuracy of the model also increases; however, using more trees than required is unnecessary and adds to the computational time (Breiman, 2001). In this study, 200 decision trees were employed. Additionally, this study adopted the approach of Thanh Noi and Kappas (2017) and Waśniewski et al. (2020) in selecting the number of features per split, utilizing the default option of the square root of the total number of features.

Accuracy assessment is an essential part of LULC classification because it evaluates the quality of classified images. Commonly used measures for evaluating the model performance include the user's accuracy, producer's accuracy, overall accuracy, and the Kappa coefficient. Overall accuracy measures the percentage of correctly classified pixels, whereas producer accuracy and user accuracy represent omission error and commission error, respectively (Olofsson et al., 2014). In GEE, all these metrics were calculated from the validation dataset. The equations for calculating overall accuracy and the Kappa coefficient are given in the Appendix.

LULC change detection

We determined LULC change by identifying the changes in the pixels of two classified raster images. For instance, this technique will identify pixels that were of the forest LULC class in 1 year and then converted to the settlement class in another year. This post-classification change detection technique is commonly used to determine the temporal transformation of LULC classes (Haque & Basak, 2017). This study used the Change Detection Wizard of ArcGIS Pro to determine the changes in LULC classes between 2016 and 2024. This Wizard can utilize raster images of numerous time frames to find pixels that have changed over time. The Categorical Change option was utilized to compute the changed and unchanged pixels for each LULC class. This analysis focused on identifying pixels in the water, forest, and agriculture/open field classes in 2016 that have been converted to the settlement class by 2024.

Fragmentation analysis

The fragmentation of forests has been identified as a major global problem for forest conservation (Reddy et al., 2013). Fragmentation analysis is frequently applied by researchers when studying the dynamics and management of forests (Kadioğulları, 2013; Midha et al., 2010; Navarro Cerrillo et al., 2019). This analysis identifies how various elements in a landscape are arranged and positioned in relation to each other,

essentially describing the spatial characteristics of the area. Three levels of metric are usually used for the analysis: patch, class, and landscape. Patch is the smallest component of the landscape and indicates a discrete area within a landscape distinguished by uniform environmental characteristics that contrast with its surrounding areas (Wiens, 1976). Patch can be delineated across various spatial scales and serves as the building block for class and landscape metrics. Class-level metrics are the combination of patches of a specific patch type, generated by aggregating patches of a specific class. The combination of class-level metrics defines the landscape-level metrics, which represent the entire landscape or study area (McGarigal et al., 2012). This study obtained class-level metrics for each LULC class and used both class-level and landscape-level metrics to study landscape fragmentation. These metrics are contingent upon scale; thus, the outcomes of these metrics are inherently affected by the resolution of the raster images.

Following the metrics used in Singh et al. (2014) and Legarreta-Miranda et al. (2021), seven metrics at class-level and two metrics at landscape-level were selected as suitable measures for this study (Legarreta-Miranda et al., 2021; Singh et al., 2014). A brief description of the class and landscape metrics that are relevant to this study is provided in Table 2. This study used FRAGSTATS (version 4.2) for fragmentation analysis, which is a software that analyzes the spatial pattern of landscape using different metrics. Classified TIFF images from 2016 and 2024 were utilized for this analysis. Between rook's contiguity (4-neighbor, only edges) and queen's contiguity (8-neighbor, edges and corners), latter was chosen to define neighborhood for identifying the relative position of the units (patch or class) in this process.

The metrics used for class level are class area (CA), aggregation index (AI), percentage of land (PLAND), landscape division index (DIVISION), number of patches (NP), patch density (PD), and patch cohesion index (COHESION). For landscape level, only number of patches (NP) and patch density (PD) metrics were used. Class area represents the total area covered by a specific patch type, such as a particular LULC class within a landscape. For instance, in the context of forest fragmentation, class area for forest patches would indicate the total area covered by the forest LULC class (McGarigal et al., 2012). The aggregation index measures the actual number of shared edges between pixels of the same patch type to the maximum number of possible shared edges. This metric is expressed as a percentage, ranging from 0 to 100, where 0 indicates no adjacencies, while a higher value indicates greater patch aggregation, and reaches 100 when the landscape consists of a single compact patch (He et al., 2000; McGarigal et al., 2012). Percentage of land represents the share of land area corresponding to a specific patch type of land-cover class, where higher values indicate a greater abundance of patches of that class within the landscape (McGarigal et al., 2012). The landscape division index assesses the likelihood of two randomly selected pixels in a landscape not belonging to the same patch of the corresponding patch type. It ranges from 0 to less than 1, where values of 0 indicate a compact landscape, while values that approach 1 indicate a fragmented landscape into numerous small patches (McGarigal et al., 2012). Number of patches indicates the count of patches present in a certain land-cover class of a landscape. It is usually greater than or equal to 1 but has no limit and can be calculated in both class and landscape scales (McGarigal et al., 2012). Patch density is calculated by dividing the

Table 2 Description of fragmentation analysis metrics (McGarigal et al., 2012)

Name of metrics	Description	Unit	Range	Level
CA	Class area of all patches of the same patch type	Hectares	CA > 0	Class
PLAND	Percentage of landscape of respective patch type	Percent	0 < PLAND ≤ 100	Class
NP	Number of patches of respective patch type	Unitless	NP ≥ 1	Class and landscape
PD	NP of respective patch type divided by total area	Number per 100 hectares	PD > 0	Class and landscape
DIVISION	1 minus CA divided by total area, squared, and summed	Proportion	0 ≤ DIVISION < 1	Class
AI	Number of contiguities of respective patch type, divided by maximum number of contiguities, multiplied by 100	Percent	0 ≤ AI ≤ 100	Class
COHESION	Connectivity of respective patch type	Unitless	0 < COHESION < 100	Class

number of patches by the total landscape area. Patch cohesion index measures the connectivity of landscape classes, calculated using patch perimeter and area, normalized by total landscape area, and ranges from 0 to 100, where a decrease in values indicates the landscape has become more fragmented (McGarigal et al., 2012). The equations for calculating class and landscape metrics are given in the Appendix.

Normalized Difference Vegetation Index (NDVI)

NDVI is a widely applied index for measuring the quantity of greenness, density and health of vegetation (Gao et al., 2020; Yengoh et al., 2015). NDVI is calculated using two segments of the electromagnetic spectrum: red light, which falls within the wavelength range of approximately 0.6–0.7 μm , and near-infrared (NIR) light, spanning from about 0.7 to 2.5 μm . Chlorophyll in plants predominantly absorbs red light, whereas near-infrared light is mostly reflected. As a result, NDVI leverages these variations in reflectance to discern higher chlorophyll levels, indicative of thriving vegetation. The establishment of refugee camps can influence the vegetation and ecosystem of the area, particularly forest cover. Calculating the NDVI enabled the detection of both lost and healthy vegetation. We calculated NDVI using Band 4 (Red) and Band 8 (NIR) of Sentinel 2 images and the equation for calculating NDVI is $(\text{NIR} - \text{RED})/(\text{NIR} + \text{RED})$. The range of NDVI is -1 to $+1$. The positive values show vegetation of different growth stages, greenness, and health, with the highest value $+1$ indicating the healthiest and fullest vegetation. Negative values usually correspond to bodies of water, whereas values close to zero (about 0.1 or lower) correspond to non-vegetative land-cover (e.g., barren rock, sand, snow). High NDVI values from about 0.6 to 0.9 are associated with dense vegetation (e.g., forests and crops at their peak growth stage) (Remote Sensing Phenology, 2018). Moderate NDVI values, around 0.2 to 0.5, indicate areas with sparse vegetation (e.g., shrubs, grasslands) or crops showing signs of maturation and health decline.

NDVI values were calculated for 2016 and 2024 to compare the changes in forest areas before and after the Rohingya refugee migration of 2017. Areas exhibiting a drastic decrease from high NDVI values should correspond to the regions where the forest class underwent a conversion to other land-uses during the 2016–2024 period. Concurrently, lower NDVI values will indicate a decline in healthy vegetation.

Camp-specific change analysis after the 2017 refugee influx

The camp-specific analysis was conducted to assess population size and land-cover changes within each camp. The camp population data, collected from the UNHCR operational data portal (UNHCR & GoB, 2024b), was used with settlement class from the LULC classification to map camp specific population distribution and estimate population density within settlement areas of each camp instead of using the total camp area. However, historical migration records provided refugee population data only for the two officially registered camps before the 2017 influx. Therefore, the percentage of settlement change was used as a proxy for the percentage of population change to analyze its relationship with the percentage of forest change. The

change was determined by subtracting the 2016 area from the 2024 area and the percentage of change was derived by multiplying the resulting value by 100.

Bivariate correlation analysis was performed to determine the relationship between camp-specific percentage of forest change and settlement change. It measures the degree of association between two quantitative variables, expressed by the correlation coefficient, ranging between -1 and $+1$, measuring the strength and direction of the relationship (Aggarwal & Ranganathan, 2016). This analysis is therefore useful for describing the association between forest and settlement change but cannot make any cause-and-effect statements. A positive correlation exists when an increase in one variable is associated with an increase in another, while a negative correlation occurs when an increase in one variable corresponds to a decrease in the other (Aggarwal & Ranganathan, 2016).

Results

Accuracy assessment results are presented in Table 3. Out of 104, 95, 122, 104 sample pixels of water, forest, agriculture/open field, and settlement, 101, 95, 120, 99 pixels were correctly classified in 2016 image, resulting in user’s accuracy 99%, 100%, 94.5%, and 98%, respectively. Similarly, producer’s accuracy was highest in 2016 for the forest class, followed by agriculture/open field, water, and settlement. In 2024, user’s accuracy and producer’s accuracy for different LULC classes varied between 92.5 and 100% and 95.8 and 99%, respectively. The overall accuracy for the

Table 3 Accuracy assessment for 2016 and 2024

	Water	Forest	Agriculture/open field	Settlement	Producer’s accuracy
Accuracy assessment 2016					
Water	101	0	2	1	97.1
Forest	0	95	0	0	100
Agriculture/open field	1	0	120	1	98.4
Settlement	0	0	5	99	95.2
User’s accuracy (%)	99	100	94.5	98	
Overall accuracy (%)	97.6				
Kappa coefficient	0.97				
Accuracy assessment 2024					
Water	95	0	1	0	99
Forest	1	92	3	0	95.8
Agriculture/open field	2	1	98	0	97
Settlement	0	0	4	96	96
User’s accuracy (%)	96.9	98.9	92.5	100	
Overall accuracy (%)	96.9				
Kappa coefficient	0.96				

2016 classified image was 97.6% and the Kappa coefficient was 0.97, whereas, in 2024, the overall accuracy was 96.9% and the Kappa coefficient was 0.96. Overall, these statistics indicate high accuracy and reliability of classified images.

Figures 3 and 4 illustrate the classified images of the study area in 2016 and 2024. Class area and percentage of land metrics from class-level fragmentation analysis revealed that the land-cover of the study area was dominated by forest (67.7%), followed by agriculture/open field (22.6%), settlement (5.8%), and water (4%) in 2016. However, settlement (40.5%) became the primary LULC type in 2024, followed by forest (30.7%), agriculture/open field (26.4%), and water (2.3%). The two most notable changes were a 54.6% decrease in forest cover from approximately 1688.8 hectares to 767.3 hectares (Table 4) and a nearly six-fold increase (598.5%) in settlement area from 144.8 to 1011.5 hectares. The decrease in class area and percentage of land in the forest class, alongside the increase in settlement class, reflects significant land-use changes following the refugee influx (Table 4). The agricultural area and open fields expanded from approximately 562.9 to 659.3 hectares (17.1%). The water class declined from 99.6 to 58 hectares (41.8%). In addition, LULC change detection technique identified the pixels that have been transformed from 2016 and 2024. Figure 6 displays the alteration of water, forest, agriculture/open field LULC classes into settlements. Approximately 695.2, 205.1, and 19.4 hectares of forest, agriculture/open field, and water were converted to settlement in the last 8 years.

Although class area and percentage of land metrics provide insight into changes in the camp area, other metrics reveal changes in the spatial arrangement. Table 4 presents the resulting value of the class-level metrics for 2016 and 2024. In 2016, the number of patches in the forest class was 663, whereas in 2024, the number of patches was 2468. As there is no upper limit to the number of patches, this massive increase in the number of patches indicates forest fragmentation. The

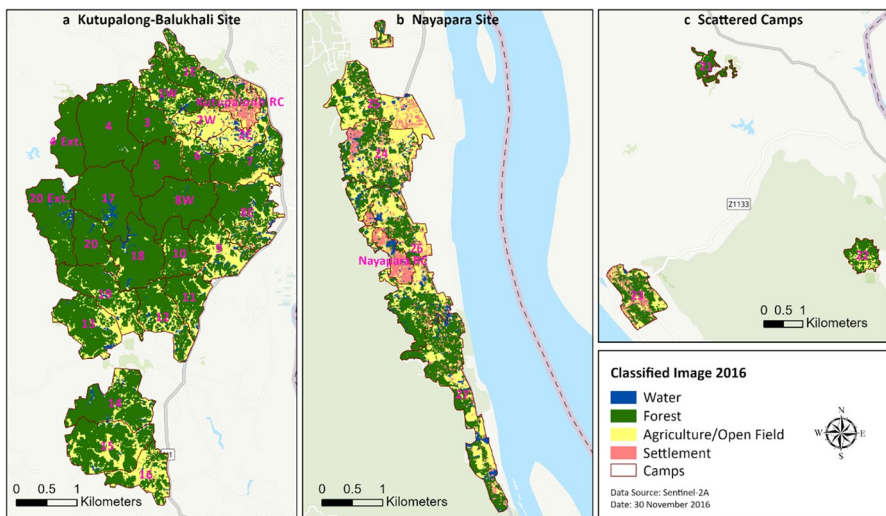


Fig. 3 Classified image of Rohingya refugee camps in 2016: **a)** Kutupalong-Balukhali site **b)** Nayapara site, and **c)** Scattered camps

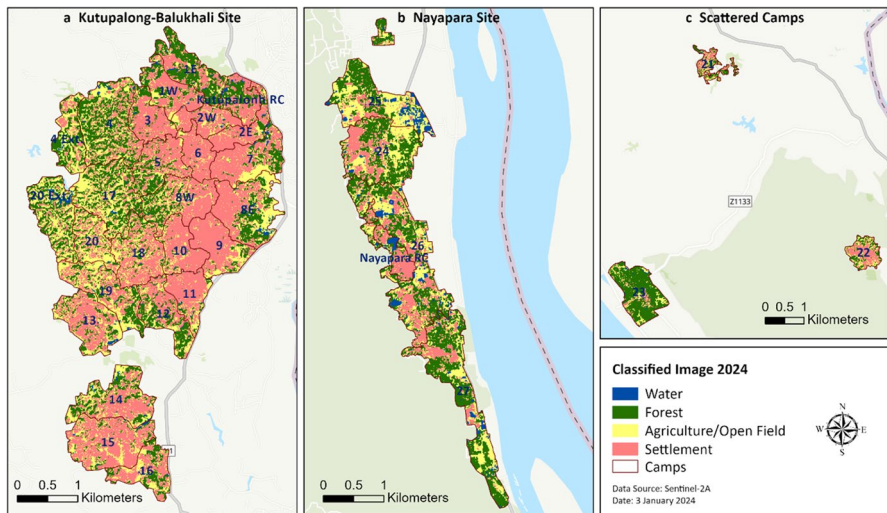


Fig. 4 Classified image of Rohingya refugee camps in 2024: **a)** Kutupalong-Balukhali site **b)** Nayapara site, and **c)** Scattered camps

increase in the number of patches in the settlement class (from 2775 to 2948) was low compared with the forest class. The agriculture/open field class also fragmented, evident from the significant increase in the number of patches from 2326 to 6500. Similarly, compared with 2016, the forest and agriculture/open field classes had very high patch density indicating a more fragmented landscape in 2024 (Table 4). The cohesion index ranges from 0 to 100, where higher values indicate less fragmentation and greater connectivity of LULC classes. This index for the settlement class increased from 85.7 to 99.1, which was the highest among classes in 2024. In addition, the cohesion index also increased for the water class (73.2 to 77.4). It shows that the settlement and water classes were more connected,

Table 4 Result of class metrics in 2016 and 2024

Class	CA	PLAND	NP	PD	COHESION	DIVISION	AI
2016							
Water	99.6	4	2049	82.1	73.2	1	49.4
Forest	1688.8	67.7	663	26.6	99.6	0.7996	91.9
Agriculture/open field	562.9	22.6	2326	93.2	96.7	0.9977	75.5
Settlement	144.8	5.8	2775	111.2	85.7	0.9999	51.5
2024							
Water	58	2.3	1,009	40.4	77.4	1	58.6
Forest	767.3	30.7	2,468	98.9	96.4	0.9968	75.2
Agriculture/open field	659.3	26.4	6,500	260.4	94.7	0.9983	60.1
Settlement	1,011.5	40.5	2,948	118.1	99.1	0.9550	79.2

whereas the forest and agriculture/open field classes are less connected at present. In addition, division index provides more clarity on land fragmentation as it uses class area and the number of patches, where 1 means highly fragmented land with numerous patches. The division index for forest increased from 0.7996 to 0.9968 and for settlement decreased from 0.9999 to 0.955. This indicates forest became more fragmented, whereas the settlement class became more contiguous. The aggregation index was 91.9 for forest in 2016; however, it decreased to 75.2 in 2024. A similar reduction, from 75.5 to 60.1, in this index value occurred for the agriculture/open field class. The aggregation index ranges between 0 and 100. A decrease in the aggregation index for these two classes indicates that these classes have become more fragmented. However, for the settlement class, the aggregation index value increased from 51.5 to 79.2, highlighting that it has become more aggregated after migration.

At the landscape-level analysis, an increase in the two metrics indicates greater fragmentation across the landscape. As shown in Fig. 5, the number of patches increased from 7724 to 12,917, while patch density rose from 309.4 to 517.5. These substantial increases in both the number of patches and patch density reveal that the entire landscape has undergone significant changes, resulting in greater fragmentation following the extensive establishment of settlements (Fig. 6).

Analysis of changes in NDVI further evaluated the impacts of migration on forest cover by spatially visualizing the areas that once had healthy vegetation but have been cleared since 2017. The dominance of dark green areas in the 2016 NDVI image indicates dense and healthy vegetation in the central and western parts of the Kutupalong-Balukhali site and extension camps 14, 15, and 16 (Fig. 7). The 2024 image (Fig. 7) shows a clear decline in areas with high NDVI values (>0.6). The

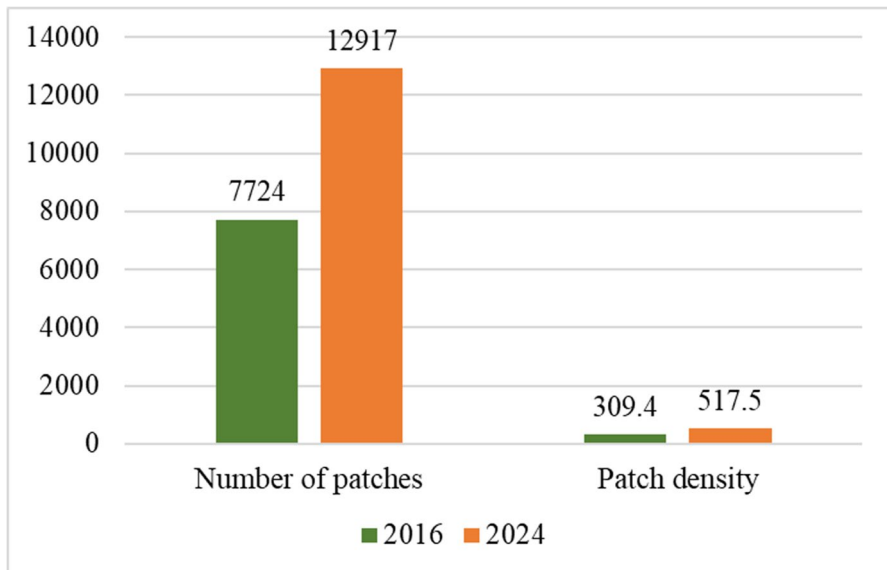


Fig. 5 Changes in number of patches and patch density generated from landscape metrics

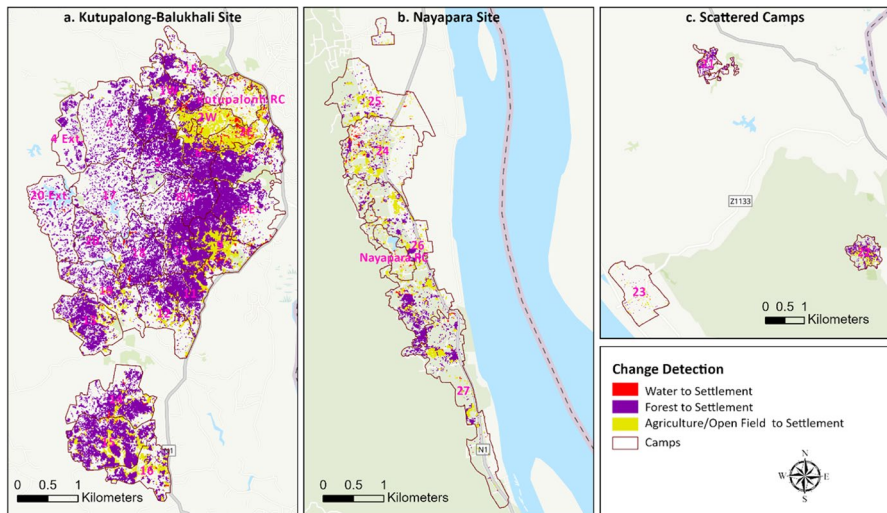


Fig. 6 Change detection map displays the transformation of different LULC classes into settlement between 2016 and 2024: **a)** Kutupalong-Balukhali site **b)** Nayapara site, and **c)** Scattered camps

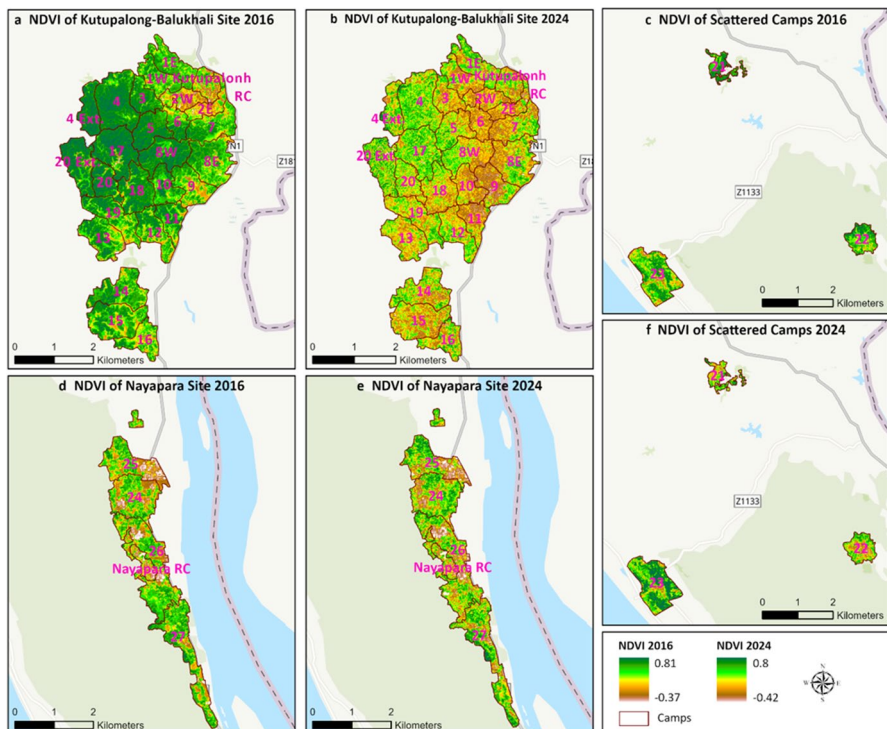


Fig. 7 NDVI analysis for the years 2016 and 2024; **a)** Kutupalong-Balukhali site in 2016, **b)** Kutupalong-Balukhali site in 2024, **c)** Scattered camps in 2016, **d)** Nayapara site in 2016, **e)** Nayapara site in 2024, and **f)** Scattered camps in 2024. Positive NDVI values correspond to healthy vegetation, with dark green color indicative of dense forest cover. Conversely, near zero and lower NDVI values represent non-vegetation land-cover

prevalence of NDVI values near or lower than zero in the 2024 image, compared to the 2016 image, validates the expansion of camp settlements. The Kutupalong-Balukhali camp has experienced the most forest loss and degradation of dense forest which is visible from Fig. 7. Closed Camp 23 and the Nayapara camp have retained most of their vegetation. The northwest and east sides of the Kutupalong-Balukhali site still have a few fragmented patches of healthy forest that can be preserved. The NDVI change map between 2024 and 2016 is displayed in Fig. 8 to visualize the most impacted areas with highest vegetation loss and portrayed those areas in red.

The correlation between camp-specific forest change and settlement change showed a negative correlation between these two variables. The correlation coefficient was -0.73 and statistically significant at the 0.05 level that indicates a strong negative linear relationship between forest change and settlement change between 2016 and 2024. This result shows that changes in forest cover are inversely related to changes in settlement. The correlation analysis indicated that Rohingya refugee camps experienced a considerable forest loss and settlement increase concurrently during this period. The results of camp specific analyses on LULC change and population information are presented in Table 5. Additionally, the Rohingya refugee population of 2024 and percentage of forest change and settlement change in camps were presented in Figs. 9 and 10, respectively.

Population analysis indicates that the highest refugee population (57,922) resides in Camp 15, followed by Camps 13, 1E, 26, and 7, each with over 40,000 refugees. Estimates suggest that Camp 1E is likely the most overcrowded camp, with 42,997 individuals residing in an area where 2324 individuals live per square hectare. For the rest of the camps, the population density varies from 559 to 1300 individuals per hectare of settlement, with 10 camps over 1000 population density.

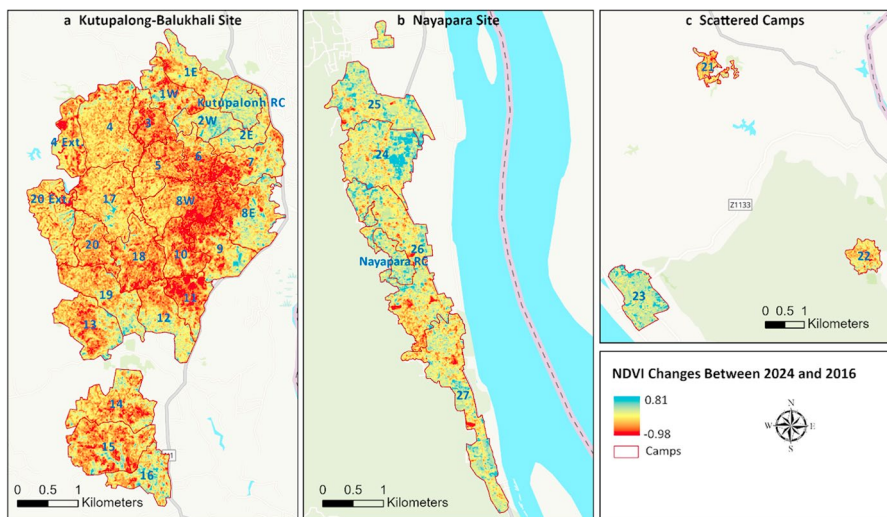


Fig. 8 NDVI Changes between 2024 and 2016: **a)** Kutupalong-Balukhali site **b)** Nayapara site, and **c)** Scattered camps. Red color regions with negative NDVI indicate the most impacted areas with highest vegetation loss

Table 5 Estimates of camp-specific population in the settlement area in 2024 and associated forest loss in camps between 2016 and 2024

Camp name	Population in 2024	Population density per hectare set- tlement area in 2024	Total camp area (hectare)	Change in settlement area (hectare)	Change in forest area (hectare)
Camp 1E	42,997	2324	63.4	16.4	-12.6
Camp 1W	39,917	1300	53.4	29.6	-26
Camp 2E	27,389	954	39.1	26.3	-2
Camp 2W	25,387	857	39.2	28.6	-2.3
Camp 3	37,576	1175	45.4	31.7	-35.1
Camp 4	34,328	1188	115.5	28.82	-61.29
Camp 4 Extension	8886	877	49.8	10.1	-30
Camp 5	27,698	867	61.5	32	-43.6
Camp 6	26,280	836	36.1	31.3	-24.8
Camp 7	40,262	952	71.4	40.8	-38.1
Camp 8E	32,405	746	95.7	39.6	-42.9
Camp 8W	33,544	793	77.2	42.3	-57.9
Camp 9	36,316	730	64.9	48.5	-31.5
Camp 10	31,743	802	49.6	39.4	-41.5
Camp 11	32,837	944	46.6	34.2	-33.6
Camp 12	29,044	1134	63.1	24.6	-21.3
Camp 13	45,550	1116	75.4	40.3	-43.7
Camp 14	35,437	927	85.7	37.4	-47.1
Camp 15	57,922	855	98.4	67.1	-62.8
Camp 16	22,598	912	52.8	23.5	-11.8
Camp 17	19,139	1121	95.4	17.1	-54
Camp 18	30,636	735	75.2	41.6	-58.5
Camp 19	27,085	1035	77	25.8	-38.7

Table 5 (continued)

Camp name	Population in 2024	Population density per hectare settlement area in 2024	Total camp area (hectare)	Change in settlement area (hectare)	Change in forest area (hectare)
Camp 20	8,610	559	48.9	15.3	-36.6
Camp 20 Extension	11,551	1018	76.6	11.3	-43.1
Camp 21	16,841	1183	40.4	14.2	-21.3
Camp 22	23,638	882	55.8	25.9	-26.8
Camp 23 (closed in 2021)	-	-	135.5	-17.3	36.3
Camp 24	26,983	843	118.1	13.3	6.8
Camp 25	9069	636	113	-0.9	3
Camp 26	41,470	852	172.1	31	-18.6
Camp 27	17,708	768	133.4	10.6	-7.1
Kutupalong RC	18,102	861	38.7	5.5	4.2
Nayapara RC	24,748	1248	32.2	0.8	3.00

*Bold figures indicate comparatively higher or lower values among camps

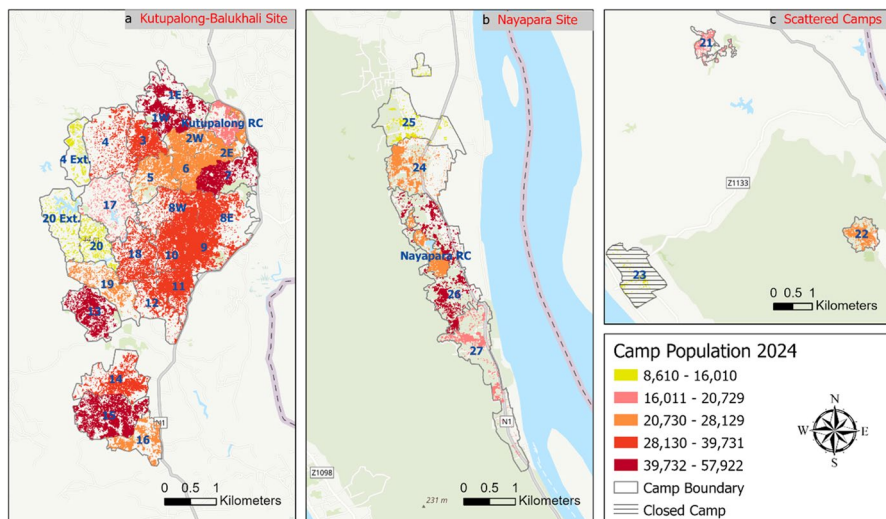


Fig. 9 Rohingya population in 2024 in the settlement areas inside camps: **a)** Kutupalong-Balukhali site, **b)** Nayapara site, and **c)** Scattered camps

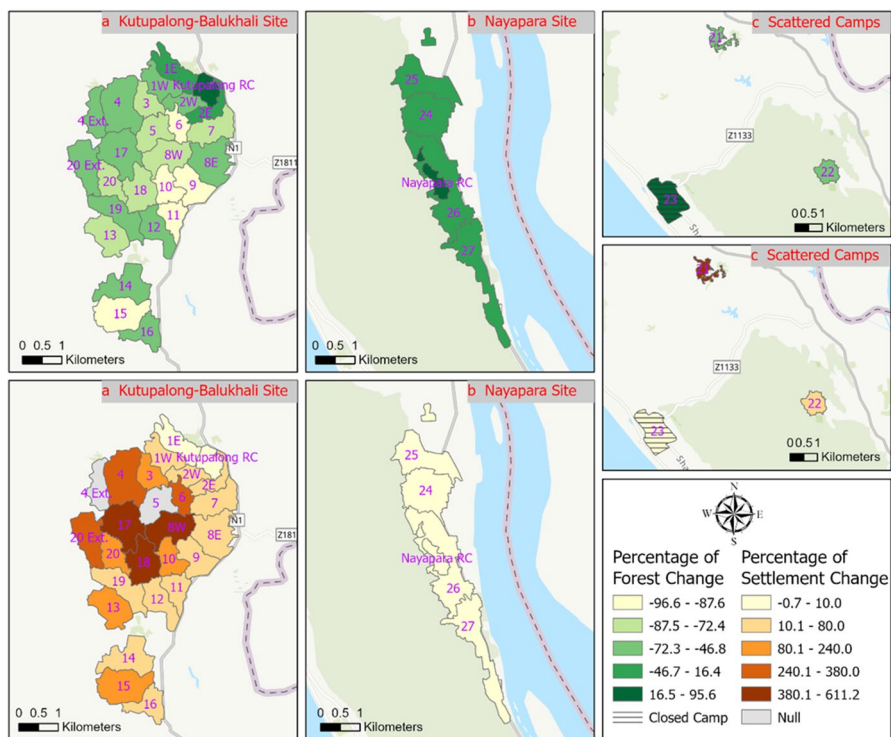


Fig. 10 Percentage of change in forest and settlement areas inside camps: **a)** Kutupalong-Balukhali site, **b)** Nayapara site, and **c)** Scattered camps

In terms of the total camp area, the largest camp is Camp 26, followed by closed Camp 23, Camps 27, 24, 4, and 25, each encompassing over 100 hectares. According to Table 5 and Fig. 10, the largest increase in the settlement area was observed in Camp 15 (67.1 hectares), which also had the highest decrease in forest area (62.9 hectares). This is the camp where the highest refugee population resides compared to other camps (Fig. 9). Additionally, Camps 7, 8W, 9, and 18 each experienced an increase of more than 40 hectares in the settlement area. Camps 6, 9, 10, 11, and 15 had more than 87% decrease in forest area from 2016 (Fig. 10). Table 5 shows that forest area decreased more than 50 hectares for Camps 4, 8W, 15, 17, and 18, whereas Camps 6, 9, 10, and 11 had lost 24.8, 31.5, 39.4, and 34.2 hectares of forest from 2016. Only the Kutupalong RC, Nayapara RC, and Camps 23, 24, and 25 had an increase in the forest area. Furthermore, Camp 23 and Camp 25 experienced a decrease in settlement area. The closed Camp 23 had the highest increase of 36.3 hectares in forest area along with the highest decrease of 17.3 hectares in settlement area. The registered camps had the least increase in settlement area with less than 5 hectares increase in forest area. The increase in forest area in these camps can be the result of the restoration efforts implemented at each camp.

Discussion and conclusion

In the analysis of the Rohingya refugee camps, the forest and water body has declined 54.6% and 41.8% respectively while settlement and agricultural land/open field has increased 598.5% and 17.1% respectively. Forest loss is consistent with the forest degradation documented in Rashid et al. (2021) and Sarker et al. (2022). The largest increase in settlement was observed in the camps of Kutupalong-Balukhali. This finding is consistent with Bappa et al. (2022) since they reported a significant increase in settlement areas from 2015 to 2021 in Kutupalong-Balukhali. The loss of the water class was also mostly observed in the Kutupalong-Balukhali site.

Approximately 695.2 hectares of forest, 205.1 hectares of agriculture/open fields, and 19.4 hectares of water, a total of 919.7 hectares of land, were transformed into settlements by 2024. The LULC analysis also indicated the conversion of a small portion of settlement areas into other land classes, notably Camp 23 and several camps at the eastern part of the Kutupalong-Balukhali and Nayapara camp clusters. This conversion is mostly observed in the closed Camp 23 where the settlement area decreased by 17.3 hectares with an increase in 36.3 hectares of forest. Some of this conversion can be the result of reforestation initiatives implemented to improve the conditions of camps and the surrounding environment. This is consistent with the findings from Hassan et al. (2023). However, analysis of long-term forest cover changes (1989–2021) from Hassan et al. (2023), alongside the timeline of major Rohingya migrations, reveals a pattern of persistent degradation in this area. Until a lasting and sustainable resolution is achieved, the risk of degradation will persist despite efforts at restoration.

The fragmentation analysis showed that the forest class had significant fragmentation, particularly at Kutupalong-Balukhali. At class-level, the increase in the number of patches, patch density, and division index and a decrease in class area,

aggregation index, patch cohesion index, and percentage of land demonstrated this fragmentation. This study found opposite results for the settlement class, which became more continuous as revealed by the increment of cohesion and aggregation index in 2024. At the landscape level, findings uncovered fragmentation of the landscape evident from an increase in both patch density and number of patches. The results and interpretation of this study for patch density, percentage of land, number of patches, and cohesion metrics are consistent with Legarreta-Miranda et al. (2021).

NDVI changes identified degraded and fragmented forest areas due to deforestation associated with settlement development, including rapid degradation in some areas of both protected forests, Inani National Park and Teknaf Wildlife Sanctuary. The central, north, and south-eastern areas of Kutupalong-Balukhali experienced severe forest loss with a decrease of over 50 hectares in existing forest area inside Camps 4, 8W, 15, 17, and 18. The analysis of changes in NDVI from this study matches the findings of Hossain and Moniruzzaman (2021).

The bivariate correlation analysis revealed that the increase of settlement areas due to migration of the Rohingya refugees was strongly associated with the decline of vegetation in the area. The population distribution indicated that the highest refugee population (57,922) resides in Camp 15 located at the center of the southern camp cluster in Kutupalong-Balukhali. This camp experienced the highest increase (67.1 hectares) in settlement area and the highest decrease (62.8 hectares) in forest area. Camps 7, 1E, and 13 of Kutupalong-Balukhali and Camp 24 of Nayapara camp clusters also have a population of more than 40,000. However, Camp 1E is identified as the most densely populated camp, followed by Camps 1W and Nayapara RC. Notably, Camps 1E, 7, and 13 exhibit both high refugee populations and high population densities. The population distribution indicated the growth of settlement areas from the pre-existing camps in the east toward the inward forest lands at the Kutupalong-Balukhali site. This matched with the UNHCR (2018) report that the makeshift settlements were growing spontaneously around the pre-existing camps in Kutupalong-Balukhali and Nayapara, expanding the camp area and forming new camps. This pattern of developing makeshift camps near the registered camps matches with the previous migration events, especially 1990s migration events, as discussed earlier in the historical account of the Rohingya migration in Bangladesh. Also, the newly arrived refugees relied on forest resources at the beginning of the massive influx as the forest provided them with materials for food and shelter before proper channels for emergency assistance were established in the expansion sites (Kamal, 2022; UNHCR, 2018). The concentration of settlements, therefore, results from this location of pre-existing camps, the proximity of the camps from Bangladesh-Myanmar international border and the availability of forest resources for building the makeshift camps and fuel (firewood) for food.

This study presents both spatial visualizations and quantifications that can aid government agencies, NGOs, and international organizations in identifying and prioritizing areas that require immediate attention. A key contribution of this research is its provision of detailed landscape fragmentation data and camp-specific LULC information, offering a comprehensive view of the current state of the camp area. Furthermore, this study tried to connect the history of Rohingya

migration with the empirical results and existing studies, which provides greater understanding of the long-term Rohingya humanitarian crisis. These findings can assist organizations involved in restoration and development efforts to assess the impact of their work more effectively and modify their action as necessary to reduce landscape fragmentation and the risks associated with rapid urbanization, including soil erosion, pollution, and rising heat levels. Initiatives should be taken toward preserving dense forests to foster their long-term regeneration into mature forests. Regular monitoring is needed for the protected forest areas located at the western edge of the camp clusters to better understand the current extent of refugees' dependency on these resources and the reason behind it. Additionally, the actions of host communities contributing to forest degradation must also be tracked and documented. Lastly, the stateless condition of the Rohingya refugees must be urgently addressed to restore their rights and alleviate the fear of further displacement. Resolving this longstanding issue is essential for allowing them to plan for their future independently of foreign aid and the restrictions imposed by host countries.

This study is limited to the camps in Cox's Bazar district, which excludes the Bhasan Char Camp in Noakhali district. The relocation of refugees to Bhasan Char and the closure of Camp 23 may have influenced the LULC and population dynamics in the study area. Furthermore, unavailability of population data for areas surrounding the registered camps before the 2017 influx limited this study to analyze the relationship between settlement change and forest change. Another limitation of this study is the presence of local people living in close proximity to refugees in some camps, which were not considered in the population analysis of settlements. Furthermore, this study conducted the class-specific fragmentation analysis but did not address the camp-specific changes in fragmentation. Also, this study investigated the spatial pattern and land fragmentation with images of 2 years and did not consider weather parameters or seasonal variations in the analysis of NDVI changes. Future research may use more images, track spatial pattern changes with NDVI time series analysis, and detect abrupt land-cover changes with the Breaks for Additive Seasonal and Trend (BFAST) method (Khan et al., 2022; Li et al., 2020). Furthermore, analyzing fragmentation within and outside camps may reveal impacts on host communities and the surrounding environment.

Appendix

Accuracy assessment

The equations for calculating overall accuracy and the Kappa coefficient are given below (Congalton, 1991).

$$\text{Overall Accuracy (\%)} = \frac{\text{Sum of correctly classified pixels}}{\text{Total number of reference pixels}} \times 100 \quad (1)$$

$$\text{Kappa Coefficient (K)} = \frac{N \sum_{i=1}^r X_{ii} - \sum_{i=1}^r (X_{i+} * X_{+i})}{N^2 - \sum_{i=1}^r (X_{i+} * X_{+i})} \quad (2)$$

Here, N = number of observations,
 r = number of rows in the error matrix,
 X_{ii} = observations in row i and column i ,
 X_{+i} = marginal total column i , and
 X_{i+} = marginal total of row i ,

Fragmentation metrics

The equations for calculating CA, AI, PLAND, DIVISION, NP, PD, and COHESION metrics are given below (McGarigal et al., 2012).

$$\text{CA} = \sum_{j=1}^n a_{ij} \left(\frac{1}{10,000} \right) \quad (3)$$

a_{ij} is area of patch ij .

$$\text{AI} = \left(\frac{g_{ii}}{\max \rightarrow g_{ii}} \right) (100) \quad (4)$$

g_{ii} is number of like joins between pixels of a certain class and $\max \rightarrow g_{ii}$ is the maximum number of joins between pixels of a certain class.

$$\text{PLAND} = P_i = \frac{\sum_{j=1}^n a_{ij}}{A} (100) \quad (5)$$

P_i is percentage of landscape of a patch type (class) i and A is total area of the landscape (m^2).

$$\text{DIVISION} = 1 - \sum_{j=1}^a \left(\frac{a_{ij}}{A} \right)^2 \quad (6)$$

$$\text{NP} = n_i \quad (7)$$

n_i represents the number of patches of specific patch type (class) of landscape.

$$\text{PD} = \frac{n_i}{A} (10,000) (100) \quad (8)$$

$$\text{COHESION} = \left| 1 - \frac{\sum_{j=1}^n p_{ij}}{\sum_{j=1}^n p_{ij} \sqrt{a_{ij}}} \right| \cdot \left| 1 - \frac{1}{\sqrt{Z}} \right|^{-1} \cdot (100) \quad (9)$$

p_{ij} is the perimeter of patch ij , a_{ij} is the area of patch ij , and Z indicates the total cells in the landscape.

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Data Source https://scihub.copernicus.eu/https://data.unhcr.org/en/country/bgd?_sm_au=iVVZ3p7tDsSn7NJq

Declarations

Competing interests The authors declare no competing interests.

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